

LIQUEO

Knowledge Discovery & Reuse

Complete Demo Package

September 30, 2026

For Financial & Business Consultants

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1. EXECUTIVE SUMMARY

Liqueo is a knowledge discovery and reuse system that enables financial and business consultants to search, discover, synthesize, and reuse insights from past engagements. The system demonstrates a complete 9-step workflow for capturing consulting knowledge and making it available for future projects.

Key Value Proposition:

- Reduce proposal development time from 3-5 days to 1 hour
- Preserve institutional knowledge through systematic capture
- Improve proposal quality by building on proven approaches
- Enable team learning through continuous knowledge reuse

2. THE PROBLEM & SOLUTION

The Challenge:

Consulting firms face a knowledge fragmentation problem. When a consultant receives an RFP for a new engagement, they:

- Don't know if similar work was done before
- Can't easily find relevant past engagements
- Must reinvent approaches from scratch
- Lose institutional learning with staff turnover

The Cost:

A fintech client RFP due Monday requires 70 hours of proposal work over the weekend. **The Solution:** Liqueo enables consultants to search for similar past engagements in seconds, analyze patterns in proven approaches, and adapt them for new contexts. What takes 70 hours from scratch takes 70 minutes with Liqueo.

3. THE 9-STEP WORKFLOW

Step 1: Identify Problem

Consultant defines the challenge with context (industry, type, constraints)

Step 2: Search Knowledge

System searches knowledge base using semantic similarity

Step 3: Identify Related Docs

System finds templates, lessons learned, success patterns

Step 4: AI Summarize

LLM provides executive summary, approach, outcomes, recommendations

Step 5: Review & Evaluate

Consultant reviews relevance and takes notes on applicability

Step 6: Select Content

Consultant chooses consulting approach, timeline, team structure to reuse

Step 7: Create New Output

Consultant creates new engagement using auto-populated template

Step 8: Tag & Classify

Add metadata (industry, type, tags) for future discovery

Step 9: Store Knowledge

New engagement saved and indexed for future reuse

4. DEMO SCENARIO (30-minute walkthrough)

Scenario Setup:

It's Friday 4pm. A fintech client sends an RFP for core banking optimization with 30% cost reduction. Proposal is due Monday 9am. Team has 70 hours to deliver.

Question: Have we solved this before? What can we learn from past engagements?

Demo Timeline:

- Part 1 (2 min): Show RFP requirement and problem
- Part 2 (3 min): Search knowledge base for similar cases
- Part 3 (4 min): View details of Investment Bank engagement (78% relevant)
- Part 4 (6 min): Show AI synthesis of patterns from 3 cases
- Part 5 (8 min): Walk through workflow Steps 5-7 (evaluate, select, create)
- Part 6 (3 min): Show completed engagement ready for proposal

Expected Outcomes:

- Find 3 similar banking cases ranked by relevance
- See full engagement details (approach, outcomes, timeline, team)
- Get AI synthesis showing recommended 3-phase approach
- Create adapted fintech proposal in 70 minutes
- Store it for next consultant facing similar challenge

5. SYSTEM FEATURES

Knowledge Base:

- Store consulting engagements with rich metadata (industry, type, value, duration)
- Tag-based classification for discovery
- Flexible filtering by multiple dimensions

Semantic Search:

- Find relevant past engagements using AI embeddings
- Keyword-based fallback (no API required)
- Relevance scoring with explanation

Recommendation Engine:

- Suggest similar engagements automatically
- Pattern matching across similar cases
- Industry-specific recommendations

LLM-Powered Synthesis:

- AI generates insights from similar cases
- Visible source traceability
- Graceful degradation without APIs

Web Interface:

- 4 main tabs (Add, Search, Recommend, Workflow)
- File upload support (PDF, Word, Excel, CSV, Text)
- URL download (OneDrive, SharePoint)
- Modal detail views for engagement exploration

6. ARCHITECTURE OVERVIEW

System Layers:

User Applications

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Liqueo Toolkit (Public API)

■■■■ Orchestration Layer (KnowledgeSynthesizer)

■■■■ Business Logic (RecommendationEngine, KnowledgeBase)

■■■■ Service Layer (EmbeddingsManager)

■■■■ Data Model (Document, SearchResult, Recommendation)

↓

External Services

■■■■ OpenAI / Anthropic APIs (embeddings, synthesis)

■■■■ Local Storage (.liqueo/ directory)

■■■■ Cache Layer (embeddings, metadata)

Data Flow:

1. **Ingestion:** Document → Validate → Store → Embed → Cache
2. **Search:** Query → Generate embedding → Compare (cosine similarity) → Rank
3. **Recommend:** Search results → Extract patterns → Generate reasoning
4. **Synthesize:** Similar cases → Format context → LLM → Generate insights

7. SAMPLE DATA (Pre-loaded for Demo)

Three Synthetic Engagements Ready to Load:

1. Investment Bank Back-Office Reorganization

Industry: Financial Services

Value: \$2.8M | Duration: 12 months

Outcome: 40% cost reduction

Approach: 3-phase (assess → design → implement)

2. Retail Bank Technology Consolidation

Industry: Financial Services

Value: \$1.5M | Duration: 9 months

Outcome: 35% IT cost reduction

Approach: Vendor assessment + solution design

3. Payment Processor Cost Optimization

Industry: Financial Technology

Value: \$0.8M | Duration: 6 months

Outcome: 28% cost reduction

Approach: Cost analysis + vendor negotiation

Load all three with: **python load_sample_data.py**

8. QUICK SETUP (5 minutes)

Installation:

```
$ git clone https://github.com/hemalp143/Liqueo.git
$ cd Liqueo
$ python3 -m venv venv
$ source venv/bin/activate
$ pip install -r requirements.txt
```

Load Sample Data:

```
$ python load_sample_data.py
```

Start Web UI:

```
$ streamlit run app.py
```

Browser opens: <http://localhost:8501>

Supported Environments:

- Local (macOS, Windows, Linux)
- Google Colab
- Docker Container
- Production Server

API Keys (Optional):

- ANTHROPIC_API_KEY (Claude embeddings/synthesis)
- OPENAI_API_KEY (OpenAI embeddings/synthesis)
- System works without keys (keyword search fallback)

9. KNOWN LIMITATIONS

Scalability: Current system handles 100-5,000 documents on single machine

Persistence: Workflow progress stored in-memory only (refreshing loses progress)

File Formats: PDF, Word, Excel, CSV, Text supported (no PowerPoint yet)

API Costs: Semantic search ~\$0.02 per document (embeddings cached)

Accuracy: LLM synthesis can hallucinate (human review required)

Multi-User: No authentication or access control yet

Analytics: No usage tracking or ROI measurement

Integration: Manual data entry (no CRM/PM tool sync)

All limitations have documented workarounds and Phase 2+ solutions.

10. PRODUCTION ROADMAP (8 Phases, 32 Weeks)

Phase 1: Foundation ■ COMPLETE

Working prototype with 9-step workflow

Phase 2: Database Persistence (4 weeks)

Add PostgreSQL, workflow resumption, audit logging

Phase 3: Vector Database (4 weeks)

Scale to 50k+ documents with FAISS or Pinecone

Phase 4: Document Processing (4 weeks)

Add PowerPoint, OCR, text chunking, auto-summarization

Phase 5: Team & Collaboration (4 weeks)

Multi-user auth, role-based access, versioning

Phase 6: Integrations (4 weeks)

Salesforce, Monday.com, OneDrive sync, Slack bot

Phase 7: Analytics (4 weeks)

Dashboard, trending, ROI calculator, team metrics

Phase 8: Governance (4 weeks)

Quality scoring, duplicate detection, approval workflows

Timeline to Production: 8 months with full feature set

11. NEXT STEPS

Immediate (Today):

- ✓ Review this demo package
- ✓ Install Liqueo locally or in Colab
- ✓ Load sample data
- ✓ Run through 30-minute demo

Week 1 (Post-Demo):

- Collect stakeholder feedback
- Identify Phase 2 priority (likely: database persistence)
- Share documentation with team

Weeks 2-4:

- Implement database persistence layer
- Add workflow session resumption
- Begin user acceptance testing

Months 2-3:

- Vector database integration
- Multi-user architecture design
- Production security audit

For Questions:

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GitHub: <https://github.com/hemalp143/Liqueo>

Branch: `claude/knowledge-discovery-reuse-e3nr1n`
